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Introduction to Filtering Theory

1. INTRODUCTION

The problem of estimating the state of a stochastic dynamical system from noisy observations taken on the state is of central importance in engineering. Interest in this problem dates back almost two centuries to the work of Gauss. Gauss was interested in determining the orbital elements of a celestial body from (many) observations and developed the technique that is known today as *least squares*. More recently, the names of Wiener and Kalman are associated with advances in estimation theory. Estimation theory had its beginnings in problems of space (astronomy, celestial mechanics), and its most recent advances are associated with modern aerospace problems.

It is not our purpose to give a historical account of the development of estimation theory. We shall outline the major mathematical methods used in estimation and credit (hopefully accurately) the development of these methods. First, however, we define the estimation problem in the context of the mathematical model we use in this book.

Consider the discrete stochastic dynamical system (Section 3.9) described by the stochastic vector difference equation

$$x_{k+1} = \varphi(x_k , t_{k+1} , t_k) + \Gamma(x_k , t_k) w_{k+1} , \qquad k = 0, 1,..., \tag{5.1}$$

where the state at t_k is x_k , an n-vector, φ is an n-vector function, Γ is

$n \times r$, and $\{w_k, k = 1,...\}$ is an r-vector, white Gaussian sequence, $w_k \sim N(0, Q_k)$. The distribution of the initial condition x_0 is assumed given, and x_0 is independent of $\{w_k\}$. We saw in Section 3.9 that the random sequence generated by (5.1) is Markov. Let discrete, noisy, m-vector observations (measurements) y_k be given by

$$y_k = h(x_k, t_k) + v_k, \qquad k = 1, 2,..., \tag{5.2}$$

where h is an m-vector function and $\{v_k, k = 1,...\}$ is an m-vector, white Gaussian sequence, $v_k \sim N(0, R_k)$, $R_k > 0$. For simplicity, $\{w_k\}$ and $\{v_k\}$ are assumed independent, and $\{w_k\}$ is independent of x_0. We note that the joint $\{x_k, y_k\}$ process is Markov.

Now let Y_l be the sequence of observations

$$Y_l = \{y_1,..., y_l\}. \tag{5.3}$$

Given a realization of the sequence of observations $\{y_1,..., y_l\}$, that is, given Y_l, the **discrete estimation** problem consists of computing an estimate of x_k based on Y_l. If $k < l$, the problem is called the **discrete smoothing** problem; if $k = l$, it is called the **discrete filtering** problem; and if $k > l$, it is called the **discrete prediction** problem.

The filtering and prediction problems are usually associated with real-time operations, in which estimates are required on the basis of observations or data available now. In a *post mortem* (after the fact) analysis, it is possible to wait for more observations to accumulate. In that case, the estimate can be improved by smoothing.

Now consider the continuous stochastic dynamical system (Section 4.1) described by the vector (Itô) stochastic differential equation (Section 4.4),

$$dx_t = f(x_t, t)\, dt + G(x_t, t)\, d\beta_t, \qquad t \geqslant t_0, \tag{5.4}$$

where x_t and f are n-vectors, G is $n \times r$, and $\{\beta_t, t \geqslant t_0\}$ is an r-vector Brownian motion process with $\mathscr{E}\{d\beta_t\, d\beta_t^{\mathrm{T}}\} = Q(t)\, dt$. [Equation (5.4) is formally equivalent to

$$dx_t/dt = f(x_t, t) + G(x_t, t)\, w_t,$$

where $\{w_t, t \geqslant t_0\}$ is a white Gaussian process, $w_t \sim N(0, Q(t))$. See Chapter 4, particularly Sections 4.1, 4.8, and 4.10.] The properties of solutions of (5.4) are summarized in Theorem 4.5. Suppose continuous observations are taken on system (5.4), of the form

$$dz_t \quad h(x_t, t)\, dt + d\eta_t, \qquad t \geqslant t_0, \tag{5.5}$$

where z_t and h are m-vectors and $\{\eta_t, t \geqslant t_0\}$ is an m-vector Brownian motion process with $\mathscr{E}\{d\eta_t\, d\eta_t^{\mathrm{T}}\} = R(t)\, dt$, $R(t) > 0$. We suppose that $\{\beta_t\}$, $\{\eta_t\}$, and x_{t_0} are independent. [Equation (5.5) is formally equivalent to

$$y_t = h(x_t, t) + v_t,$$

with the identifications

$$y_t \sim dz_t/dt, \qquad v_t \sim d\eta_t/dt,$$

and with $\{v_t, t \geqslant t_0\}$ a white Gaussian noise process, $v_t \sim N(0, R(t))$.] Equations (5.4) and (5.5), taken together, constitute a vector (Itô) stochastic differential equation.

Suppose a realization

$$Y_\tau = \{z_s, t_0 \leqslant s \leqslant \tau\} \tag{5.6}$$

is given. The problem of estimating x_t, given Y_τ (based on Y_τ), is the **continuous estimation** problem. If $t < \tau$, the problem is called the **continuous smoothing** problem; if $t = \tau$, it is the **continuous filtering** problem; and if $t > \tau$, it is called the **continuous prediction** problem.

In addition to the discrete and continuous problems, we shall also study the mixed **continuous-discrete estimation** problem. In this problem, the state x_t evolves according to the stochastic differential equation (5.4), and observations (5.2) are taken at discrete time instants. The smoothing, filtering, and prediction problems are defined in an analogous way.

Important special cases of the problems we have defined result when the dynamics and observations are *linear*. In the discrete problem, this means that the function φ is linear in x_k :

$$\varphi(x_k, t_{k+1}, t_k) = \Phi(t_{k+1}, t_k)\, x_k$$

(Φ an $n \times n$ matrix), the matrix function Γ is independent of x_k, and the measurement function h is linear in x_k :

$$h(x_k, t_k) = M(t_k)\, x_k$$

(M an $m \times n$ matrix). Similarly, in the continuous problem,

$$f(x_t, t) = F(t)\, x_t$$

(F an $n \times n$ matrix), G is independent of x_t, and

$$h(x_t, t) = M(t)\, x_t.$$

In the linear problems, the stochastic processes involved are Gaussian

(see Sections 3.9 and 4.4) as well as Markov, and as a result the linear problems admit a particularly tractable solution.

Discrete (and continuous-discrete) estimation theory has its beginnings in the work of Gauss, which we have already mentioned. Differential correction techniques, based on the idea of least squares, have been used widely in orbit determination. The interested reader can find an account of this aspect of estimation theory in Deutsch [16]. This earlier work does not place the estimation problem in a probabilistic framework as we have done. Rather, the problem is looked upon as a deterministic problem of minimizing errors. This will be further pursued in Section 3.

Continuous estimation theory begins with Wiener [42]. Wiener solves the linear, stationary problem (F, G, M, Q, and R constant) using frequency domain techniques. His model is a probabilistic one, like ours, but the stochastic processes involved are not modeled by stochastic differential equations. The continuous filtering problem is reduced to the solution of an integral equation (the Wiener–Hopf equation). Wiener [43] gives a general, practical method of solution in the linear-stationary case, but the nonlinear and nonstationary problems remain essentially unsolved. See the text by Laning and Battin [24], for example, for further details and additional bibliography. See also Parzen [29] for an excellent summary and a new approach to Wiener's problem.

The general linear (nonstationary) filtering and prediction problem is essentially completely solved in the pioneering work of Kalman [19], Kalman and Bucy [20], and Kalman [21]. The parallel work of Stratonovich [37, 38] and Kushner [22, 23] provides the basis for subsequent developments in nonlinear filtering and prediction theory. These authors adopt the probabilistic approach that is introduced in Section 2, in which the problem is modeled via stochastic difference and differential equations, as we have done in the foregoing. We pursue this approach in Section 2.

This book deals almost exclusively with the problems of filtering and prediction. Smoothing, for linear problems, is briefly treated in Examples 7.8 and 7.16. Additional material on smoothing may be found in Bryson and Frazier [7], Rauch [31], Rauch *et al.* [32], Cox [12, 13], Friedland and Bernstein [17], and Meditch [26].

2. PROBABILISTIC APPROACH

Consider, for concreteness, the continuous filtering problem defined in the preceding section. It is clear that the conditional probability density function of x_t *given* Y_t, which we write as

$$p(x, t \mid Y_t), \tag{5.7}$$

is the complete solution of the filtering problem. This is simply because $p(x, t \mid Y_t)$ embodies all the statistical information about x_t which is contained in the available observations *and* in the initial condition $p(x, t_0)$. We are here adopting the so-called *Bayesian* point of view in that we are taking into account *a priori* (initial) data or information about x_t which is contained in $p(x, t_0)$. (See the definition of a conditional density function in Section 2.5, and also Section 3.5. The conditional density is defined in terms of the *joint* density.) Similarly, the complete solution of the continuous prediction problem is $(t > \tau)$

$$p(x, t \mid Y_\tau). \tag{5.8}$$

Analogous statements hold for the discrete and continuous-discrete filtering and prediction problems.

In the case of linear filtering and prediction problems, the densities in (5.7) and (5.8) are Gaussian (see Sections 3.9 and 4.4 and Theorem 2.13). As a result, they are characterized by their (respective) mean vectors and covariance matrices. It is then a relatively simple matter to compute these conditional densities. In the nonlinear case, the situation is vastly more difficult. In general, there does not exist a finite set of parameters which characterizes these densities. We might say that in the linear case the filter state is finite, consisting of the mean and covariance matrix. In the nonlinear case, the filter state is infinite—the whole function $p(x, t \mid Y_t)$.

With the densities (5.7) and (5.8) in hand, there still remains the question of what the estimate of the state should be. Several obvious possibilities suggest themselves, as for example, the mean, the mode (peak), the median, and so on. We need a criterion that will permit us to compare various possible estimates, and to choose the best (optimal) one.

Let $\hat{x}_t$ be an estimate of x_t given Y_τ, $t \geqslant \tau$. Then the error in the estimate is simply

$$\tilde{x}_t \equiv x_t - \hat{x}_t\,. \tag{5.9}$$

A good estimate should clearly have the property of producing small errors, or rather a small statistical measure of the errors. We proceed to define such a measure.

Let $\rho(\xi)$ be a real-valued, nonnegative, convex[1] function of the real

[1] $\rho(\xi)$ is *convex* if "straight-line interpolation" overestimates ρ. More precisely, at every ξ^0,

$$\lambda\rho(\xi) + (1 - \lambda)\rho(\xi^0) \geqslant \rho(\lambda\xi + (1 - \lambda)\xi^0),$$

$0 \leqslant \lambda \leqslant 1$, for all ξ.

n-vector ξ. We define a *loss* or criterion function $L(\tilde{x}_t)$ as any real-valued function with the properties

$$L(0) = 0, \qquad \rho(\xi^2) \geqslant \rho(\xi^1) \geqslant 0 \Rightarrow L(\xi^2) \geqslant L(\xi^1) \geqslant 0. \tag{5.10}$$

As an example, an admissible function ρ is

$$\rho(\xi) = |\, \xi \,| = (\xi^{\mathrm{T}}\xi)^{1/2}. \tag{5.11}$$

ρ clearly measures the distance from the origin, and, in view of (5.10), the loss function is nondecreasing with this distance. Note that L itself can, but need not be, convex. In the scalar case, a subclass of loss functions with properties (5.10) is the class of symmetric functions

$$L(\xi) = L(-\xi) \geqslant 0, \qquad \xi^2 \geqslant \xi^1 \geqslant 0 \Rightarrow L(\xi^2) \geqslant L(\xi^1). \tag{5.12}$$

Take the ρ in (5.11). Clearly,

$$L(\xi) = \xi^{\mathrm{T}} S \xi, \tag{5.13}$$

where the $n \times n$ (symmetric) matrix S is positive semidefinite, has properties (5.10).

Having defined a reasonable class of loss functions, we seek that estimate $\hat{x}_t$ of x_t which minimizes the average or *expected loss*

$$\mathscr{E}\{L(\tilde{x}_t)\}.$$

For a certain class of density functions $p(x, t \mid Y_\tau)$, the answer is contained in the following theorem due to Sherman [34], which we state here without proof. See also Sherman [35] for a discussion of loss functions (5.12).

Theorem 5.1 (Sherman). *Let x be a random vector with mean μ and density function $p(x)$. Let $L(\tilde{x})$, $\tilde{x} = x - \hat{x}$, be of class* (5.10). *If $p(x)$ is symmetric about μ, and unimodal,*[2] *then $\hat{x} = \mu$ minimizes $\mathscr{E}\{L(\tilde{x})\}$.*

Now the expected loss can be written ($t \geqslant \tau$) as

$$\mathscr{E}\{L(\tilde{x}_t)\} = \mathscr{E}\{\mathscr{E}\{L(\tilde{x}_t) \mid Y_\tau\}\}. \tag{5.14}$$

Since $\mathscr{E}\{L(\tilde{x}_t) \mid Y_\tau\}$ depends only on Y_τ (is independent of $\hat{x}_t$), $\mathscr{E}\{L(\tilde{x}_t)\}$

[2] $p(x)$ is *unimodal* if it has only one "peak." More precisely, if $p(x)$ is symmetric about μ, then $p(x)$ is unimodal if the distribution function $F(x)$ is convex for $x_i \leqslant \mu_i$.

is minimized by minimizing $\mathscr{E}\{L(\tilde{x}_t) | Y_\tau\}$. Applying Theorem 5.1 with $p(x, t \mid Y_\tau)$, we have

Theorem 5.2. *Let $p(x, t \mid Y_\tau)$ have mean $\hat{x}_t^\tau \triangleq \mathscr{E}\{x_t \mid Y_\tau\}$. Suppose $p(x, t \mid Y_\tau)$ is symmetric about $\hat{x}_t^\tau$ and unimodal. Let $L(\tilde{x}_t)$ be a class* (5.10). *Then the optimal estimate of x_t in the continuous, discrete, and continuous-discrete filtering and prediction problems, in the sense of minimizing $\mathscr{E}\{L(\tilde{x}_t)\}$, is the conditional mean $\hat{x}_t^\tau$.*

The hypotheses of Theorem 5.2 are clearly satisfied if $p(x, t \mid Y_\tau)$ is Gaussian. We therefore have

Corollary. *Let $L(\tilde{x}_t)$ be of class* (5.10). *Then the optimal estimate of x_t in linear filtering and prediction problems (in the sense of minimizing $\mathscr{E}\{L(\tilde{x}_t)\}$) is the conditional mean $\hat{x}_t^\tau \triangleq \mathscr{E}\{x_t \mid Y_\tau\}$.*

Examples of loss functions of class (5.10) include

$$L_1(\xi) = \xi^{\mathrm{T}} S \xi, \tag{5.13}$$

$$L_2(\xi) = \sum_i c_i \mid \xi_i \mid, \qquad c_i > 0,^3 \tag{5.15}$$

$$L_3(\xi) = \begin{cases} \xi^{\mathrm{T}} S \xi & \xi^{\mathrm{T}} S \xi \leqslant c \\ c, & \xi^{\mathrm{T}} S \xi > c,^3 \end{cases} \tag{5.16}$$

and, in the scalar case [class (5.12)],

$$L_4(\xi) = c\xi^4, \qquad c > 0,^4 \tag{5.17}$$

$$L_5(\xi) = c(1 - e^{-\xi^2}), \qquad c > 0,^4 \tag{5.18}$$

and so on. Additional loss functions are sketched in Fig. 5.1. Three of them are of class (5.12) (symmetric), whereas one is asymmetric.[5]

We see that the conditional mean provides the minimum expected loss for a large class of filtering and prediction problems, and for many loss functions. Clearly, if $p(x, t \mid Y_\tau)$ is available, then the optimal estimate can be computed for any loss function.

A particularly tractable loss function is the quadratic form (5.13)

$$L_1(\xi) = \xi^{\mathrm{T}} S \xi, \tag{5.19}$$

[3] From Cox [12].

[4] From Kalman [19].

[5] Brown [6] shows that, in the case of scalar, Gaussian processes, the conditional mean is also optimal for asymmetric loss functions that are (i) nonnegative, (ii) nondecreasing for $\xi \geqslant 0$, and (iii) nonincreasing for $\xi \leqslant 0$.

where $S \geqslant 0$. The estimate that minimizes $\mathscr{E}\{L_1(\tilde{x}_t)\}$ is called (for obvious reasons) the **minimum variance** or **minimum mean square error** estimate. It turns out that the conditional mean is the minimum variance estimate for *all* filtering and prediction problems, regardless of the properties of the conditional density function.

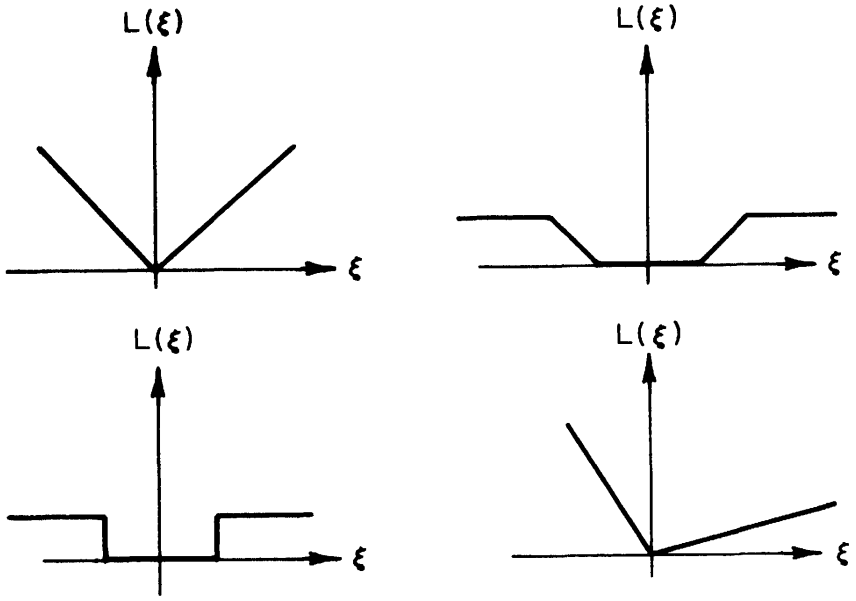


FIG. 5.1. Loss functions of class (5.10).

Theorem 5.3. *Let the estimate be a functional on Y_τ. Then the minimum variance estimate is the conditional mean.*

PROOF: Write

$$\mathscr{E}\{L_1\} = \mathscr{E}\{\mathscr{E}\{(x_t - \mu_t + \mu_t - \hat{x}_t)^{\mathrm{T}} S(x_t - \mu_t + \mu_t - \hat{x}_t)|\ Y_\tau\}\}, \quad (5.20)$$

and let

$$\mu_t = \mathscr{E}\{x_t \mid Y_\tau\}.$$

Since

$$\begin{aligned}\mathscr{E}\{(x_t - \mu_t)^{\mathrm{T}} S(\mu_t - \hat{x}_t)|\ Y_\tau\} &= (\mu_t - \hat{x}_t)^{\mathrm{T}} S\mathscr{E}\{(x_t - \mu_t)|\ Y_\tau\} \\ &= (\mu_t - \hat{x}_t)^{\mathrm{T}} S(\mu_t - \mu_t) = 0\end{aligned}$$

(μ_t and the estimate $\hat{x}_t$ are constant with respect to expectation conditioned on Y_τ), and since

$$\mathscr{E}\{(x_t - \mu_t)^{\mathrm{T}} S(x_t - \mu_t)\}$$

is independent of $\hat{x}_t$, (5.20) becomes

$$\mathscr{E}\{L_1\} = \text{const} + \mathscr{E}\{(\mu_t - \hat{x}_t)^{\mathrm{T}} S(\mu_t - \hat{x}_t)\}. \quad (5.21)$$

Since S is positive semidefinite, (5.21) is clearly minimized by setting $\hat{x}_t = \mu_t$. ■

We note that the conditional mean is an *unbiased estimate*, that is,

$$\mathscr{E}\{(x_t - \hat{x}_t^\tau)\} = \mathscr{E}\{x_t\} - \mathscr{E}\{\mathscr{E}\{x_t \mid Y_\tau\}\} = 0.$$

From the results of this section, it is clear that the first order of business in solving filtering and prediction problems is the determination of the conditional probability density function $p(x, t \mid Y_\tau)$. This is essentially what we mean by the probabilistic approach. With the conditional density in hand, the optimal estimate can be computed for any loss function. However, in view of Theorem 5.2, and especially Theorem 5.3, our second order of business will be the determination of the conditional mean. This is the minimum variance estimate, and our concern will be primarily with this estimate. What we seek are equations of evolution of the conditional density and the conditional mean. We have already developed such equations in the absence of observations. (See Section 3.9 in the discrete case and Section 4.9 in the continuous case.) These equations will constitute the optimal filter and/or predictor.

We will find that the conditional mean, which is the first moment of the conditional density function, in general depends on the higher-order moments. Of course, in linear problems only second moments are involved. These are contained in the *conditional covariance matrix*

$$P_t^\tau \triangleq \mathscr{E}\{(x_t - \hat{x}_t^\tau)(x_t - \hat{x}_t^\tau)^{\mathrm{T}} \mid Y_\tau\},$$

and we will have to develop an equation of evolution for this matrix. Since

$$\mathscr{E}\{(x_t - \hat{x}_t^\tau)^{\mathrm{T}}(x_t - \hat{x}_t^\tau)\} = \operatorname{tr} \mathscr{E}\{P_t^\tau\},$$

P_t^τ provides a measure of the goodness of the estimate. It can be argued that knowledge of P_t^τ is just as important as knowing the estimate $\hat{x}_t^\tau$ itself. An estimate is meaningless unless one knows how good it is.

3. STATISTICAL METHODS

Consider the discrete estimation problem defined in Section 1. Instead of regarding w_k and v_k in (5.1) and (5.2) as random inputs with well-defined statistics, consider them simply as errors of unknown character. Equation (5.1) is no longer a stochastic difference equation, but rather an ordinary difference equation that could be solved if the

errors were known. w_k might represent a deterministic error in modeling the dynamical system via the difference equation (5.1). Suppose some *a priori* estimate of x_0, call it $\bar{x}_0$, is available. Suppose noisy observations $\{y_1, \ldots, y_N\}$ are given by (5.2). We wish to determine (estimate) $\{x_0, \ldots, x_N\}$ so that, simultaneously, the errors in the dynamical system and in the observations are small. Roughly speaking, we want to pass the solution of (5.1), as closely as possible, through the observations.

Taking the classical *least squares* approach, it is appropriate to minimize, with respect to $\{x_0, \ldots, x_N ; w_1, \ldots, w_N\}$, the function

$$J_N = \frac{1}{2}(x_0 - \bar{x}_0)^{\mathrm{T}} P_0^{-1}(x_0 - \bar{x}_0) + \frac{1}{2}\sum_{k=1}^{N}(y_k - h(x_k, t_k))^{\mathrm{T}} R_k^{-1}(y_k - h(x_k, t_k))$$

$$+ \frac{1}{2}\sum_{k=1}^{N} w_k{}^{\mathrm{T}} Q_k^{-1} w_k, \tag{5.22}$$

subject to the constraints

$$x_{k+1} = \varphi(x_k, t_{k+1}, t_k) + \Gamma(x_k, t_k)\, w_{k+1}, \qquad k = 0, 1, \ldots, N-1. \tag{5.23}$$

The appearance of the first term in J_N reflects our belief in the prior estimate. The positive definite matrices P_0^{-1}, R_k^{-1}, and Q_k^{-1} are here to be regarded as *weighting* matrices, which are quantitative measures of our belief in the prior estimate, the observation model, and the dynamical system model, respectively. In practice, they are "fiddling parameters" that help us obtain a good least squares fit.

Similar considerations in the continuous estimation problem lead to the minimization, with respect to $\{x_\tau, t_0 \leqslant \tau \leqslant t\}$ and $\{w_\tau, t_0 \leqslant \tau \leqslant t\}$, of the functional

$$J_t = \frac{1}{2}(x_{t_0} - \bar{x}_{t_0})^{\mathrm{T}} P_{t_0}^{-1}(x_{t_0} - \bar{x}_{t_0})$$

$$+ \frac{1}{2}\int_{t_0}^{t}(y_\tau - h(x_\tau, \tau))^{\mathrm{T}} R^{-1}(\tau)(y_\tau - h(x_\tau, \tau))\, d\tau$$

$$+ \frac{1}{2}\int_{t_0}^{t} w_\tau{}^{\mathrm{T}} Q^{-1}(\tau)\, w_\tau\, d\tau, \tag{5.24}$$

subject to the constraint

$$dx_\tau/d\tau = f(x_\tau, \tau) + G(x_\tau, \tau)\, w_\tau, \qquad t_0 \leqslant \tau \leqslant t. \tag{5.25}$$

To be specific, consider the discrete problem in (5.22) and (5.23). The solution of that problem, the minimizing sequence $\{\bar{x}_0, \ldots, \bar{x}_N\}$,

is the desired smoothing solution. $\bar{x}_N$ is the filtering solution. Prediction (to t_M, say) can be accomplished by adding

$$\frac{1}{2}\sum_{k=N+1}^{M-1} w_k^{\mathrm{T}} Q_k^{-1} w_k \tag{5.26}$$

to J_N. It is easy to see that the minimizing sequence $\{w_{N+1},\ldots, w_{M-1}\}$ is

$$w_k \equiv 0, \qquad k = N+1,\ldots, M-1.$$

As a result, prediction is accomplished by assuming an errorless dynamical system:

$$y_{k+1} = \varphi(y_k, t_{k+1}, t_k); \qquad k = N, N+1,\ldots; \qquad y_N = \bar{x}_N. \tag{5.27}$$

Now the minimization of J_N obviously cannot be accomplished until all observations $\{y_1,\ldots, y_N\}$ have been collected. When the additional observation y_{N+1} is obtained, a new problem, J_{N+1}, has to be solved. Proceeding this way, it soon becomes painfully clear that observations accumulate *ad infinitum*. Furthermore, the computations involved in solving the least squares minimization problem grow with N.

One way to avoid this growth in memory and computations is to separate the observations into batches,

$$\{y_1,\ldots, y_N\}, \{y_{N+1},\ldots, y_{2N}\},\ldots,$$

and to solve the least squares problem separately for each batch of observations. This procedure is sometimes called *batch processing*. Information from one batch can be carried over to the next via the first term in J_N. That is, information about x_0 from previous batches can be considered as *a priori* information for the current batch. This is clearly, in general, an *ad hoc* procedure and is not equivalent to processing all the data simultaneously.

Another procedure, called *recursive least squares*, consists of solving the problem J_{N+1} in terms of the observation y_{N+1} and $\bar{x}_N$ (the filtering solution for problem J_N). The goal is to obtain a difference equation for the estimate $\bar{x}_N$ with the current observation as a forcing term. This would lead to a filter similar in structure to that described in Section 2. It turns out that a recursive least squares solution can be obtained for linear problems. Approximate recursive solutions are available in the nonlinear case. The recursive least squares approach was actually inspired by probabilistic results that automatically produce an

equation of evolution for the estimate (the conditional mean). In fact, much of the recent least squares work did nothing more than rederive the probabilistic results (perhaps in an attempt to understand them). As a result, much of the least squares work contributes very little to estimation theory. Notable exceptions include work in smoothing. Smoothing solutions were first obtained via the least squares approach.

The least squares approach that we developed above has no probabilistic meaning. We simply set out to minimize errors in a deterministic problem. By reinterpreting the errors $\{w_k\}$ and $\{v_k\}$ as random sequences with statistics defined in Section 1, we shall see that the discrete least squares problem in (5.22), (5.23) is equivalent to maximizing the conditional probability density function

$$p(x_0, \ldots, x_N \mid y_1, \ldots, y_N) \tag{5.28}$$

with respect to $\{x_0, \ldots, x_N\}$, provided Γ in (5.1) is a time function $\Gamma(t_k)$, independent of x_k.[6] A similar interpretation will be made in the continuous problem. Our development follows Cox [12, 13].

Using Bayes' rule, we write the density in (5.28) as

$$p(x_0, \ldots, x_N \mid y_1, \ldots, y_N) = \frac{p(y_1, \ldots, y_N \mid x_0, \ldots, x_N)\, p(x_0, \ldots, x_N)}{p(y_1, \ldots, y_N)}. \tag{5.29}$$

Note that only the numerator in (5.29) depends on $\{x_0, \ldots, x_N\}$. Now with $\{x_0, \ldots, x_N\}$ given, $\{y_1, \ldots, y_N\}$ [see (5.2)] are independent, since the $\{v_k\}$ sequence is white. Therefore, applying Theorem 2.7, we get

$$p(y_1, \ldots, y_N \mid x_0, \ldots, x_N) = \prod_{k=1}^{N} p_{v_k}(y_k - h(x_k, t_k)). \tag{5.30}$$

Since the sequence generated by (5.1) is Markov, we have

$$\begin{aligned} p(x_0, \ldots, x_N) &= p(x_0) \prod_{k=1}^{N} p(x_k \mid x_{k-1}) \\ &= p(x_0) \prod_{k=1}^{N} p_{\Gamma w_k}(x_k - \varphi(x_{k-1}, t_k, t_{k-1})). \end{aligned} \tag{5.31}$$

[6] See Friedland and Bernstein [17] for the case in which Γ is state dependent. We note that Bryson and Frazier [7] and Cox [13] contain an error in that the state dependence of Γ (G in the continuous problem) is there neglected.

The last equality in (5.31) follows by Theorem 2.7 in view of the fact that the $\{w_k\}$ sequence is white. Collecting these results, we have

$$p(x_0, \ldots, x_N \mid y_1, \ldots, y_N) = cp(x_0) \prod_{k=1}^{N} p_{v_k}(y_k - h(x_k, t_k)) \times \prod_{k=1}^{N} p_{\Gamma w_k}(x_k - \varphi(x_{k-1}, t_k, t_{k-1})), \tag{5.32}$$

where c is a constant independent of $\{x_0, \ldots, x_N\}$.

Let us assume that the initial condition $x_0 \sim N(\bar{x}_0, P_0)$, $P_0 > 0$. Recall that $v_k \sim N(0, R_k)$, $R_k > 0$, and $w_k \sim N(0, Q_k)$, $Q_k \geqslant 0$, so that

$$\Gamma w_k \sim N(0, \Gamma Q_k \Gamma^{\mathrm{T}}).$$

We assume for simplicity (see Section 3.9) that $\Gamma Q_k \Gamma^{\mathrm{T}} > 0$ and $Q_k > 0$; both Γ and Q_k are $n \times n$ and nonsingular. Then (5.32) becomes

$$\begin{aligned} &p(x_0, \ldots, x_N \mid y_1, \ldots, y_N) \\ &\quad = c' \exp \Big\{ -\tfrac{1}{2}(x_0 - \bar{x}_0)^{\mathrm{T}} P_0^{-1}(x_0 - \bar{x}_0) \\ &\qquad - \frac{1}{2} \sum_{k=1}^{N} (y_k - h(x_k, t_k))^{\mathrm{T}} R_k^{-1}(y_k - h(x_k, t_k)) \\ &\qquad - \frac{1}{2} \sum_{k=1}^{N} (x_k - \varphi(x_{k-1}, t_k, t_{k-1}))^{\mathrm{T}} (\Gamma Q_k \Gamma^{\mathrm{T}})^{-1} (x_k - \varphi(x_{k-1}, t_k, t_{k-1})) \Big\}, \end{aligned} \tag{5.33}$$

where c' is a constant independent of $\{x_0, \ldots, x_N\}$. Maximizing (5.33) with respect to $\{x_0, \ldots, x_N\}$ is equivalent to minimizing

$$\begin{aligned} &\frac{1}{2}(x_0 - \bar{x}_0)^{\mathrm{T}} P_0^{-1}(x_0 - \bar{x}_0) + \frac{1}{2} \sum_{k=1}^{N} (y_k - h(x_k, t_k))^{\mathrm{T}} R_k^{-1}(y_k - h(x_k, t_k)) \\ &\quad + \frac{1}{2} \sum_{k=1}^{N} (x_k - \varphi(x_{k-1}, t_k, t_{k-1}))^{\mathrm{T}} (\Gamma Q_k \Gamma^{\mathrm{T}})^{-1} (x_k - \varphi(x_{k-1}, t_k, t_{k-1})), \end{aligned} \tag{5.34}$$

which, in turn, is equivalent to minimizing J_N (5.22) with respect to $\{x_0, \ldots, x_N ; w_1, \ldots, w_N\}$, subject to constraints (5.23).

We now consider the continuous least squares problem (5.24), (5.25).

Our approach is to construct an analog of the conditional probability density function (5.28) in the continuous case. We first discretize (5.25), construct the required probability density functions, and then pass to the limit. Our development is formal.

Assume, for simplicity, that $t_0 = 0$, $Q(t) > 0$, $G(t)Q(t)G^{\mathrm{T}}(t) > 0$; Q and G are $n \times n$ and nonsingular. Partition $[0, t]$

$$0, \delta\tau, \ldots, (N-1)\,\delta\tau = t,$$

with $\delta\tau > 0$ and small. Then the discretized dynamics and observations are

$$\begin{aligned} x_{(k+1)\delta\tau} &= x_{k\delta\tau} + f(x_{k\delta\tau}, k\,\delta\tau)\,\delta\tau + G(k\,\delta\tau)\,w_{k\delta\tau}\,\delta\tau, \\ y_{k\delta\tau} &= h(x_{k\delta\tau}, k\,\delta\tau) + v_{k\delta\tau}, \end{aligned} \tag{5.35}$$

where now $w_{k\delta\tau} \sim N(0, Q(k\,\delta\tau)/\delta\tau)$, $v_{k\delta\tau} \sim N(0, R(k\,\delta\tau)/\delta\tau)$ (see Section 3.8, Example 3.20).

In view of (5.29), we have

$$p(x_0, x_{\delta\tau}, \ldots, x_t \mid y_0, y_{\delta\tau}, \ldots, y_t) = \frac{p(y_0, \ldots, y_t \mid x_0, \ldots, x_t)\,p(x_0, \ldots, x_t)}{p(y_0, \ldots, y_t)}. \tag{5.36}$$

Since only the numerator in (5.36) depends on x_τ, we shall be concerned only with maximizing

$$p(y_0, \ldots, y_t \mid x_0, \ldots, x_t)\,p(x_0, \ldots, x_t). \tag{5.37}$$

Now, in view of (5.30),

$$\begin{aligned} p(y_0, \ldots, y_t \mid x_0, \ldots, x_t) &= \prod_{k=0}^{N-1} p_{v_{k\delta\tau}}(y_{k\delta\tau} - h(x_{k\delta\tau}, k\,\delta\tau)) \\ &= c_1 \exp\left\{-\frac{1}{2}\sum_{k=0}^{N-1} (y_{k\delta\tau} - h)^{\mathrm{T}} R^{-1}(k\,\delta\tau)(y_{k\delta\tau} - h)\,\delta\tau\right\}. \end{aligned} \tag{5.38}$$

If we attempt to pass to the $\delta\tau = 0$ ($N = \infty$) limit in (5.38), the normalizing constant c_1 will go to zero. To avoid this difficulty, we utilize the concept of a *probability density functional.*[7] We divide (5.38) by

$$\prod_{k=0}^{N-1} p_{v_{k\delta\tau}}(0) = c_1,$$

[7] For a rigorous treatment of this concept, see Parzen [30].

which is independent of x_τ, and *define* the probability density functional

$$p[y_\tau, \tau \in [0, t] | x_\tau, \tau \in [0, t]]$$

$$\triangleq \lim_{\substack{\delta\tau \to 0 \\ N \to \infty}} \frac{p(y_0, \ldots, y_t \mid x_0, \ldots, x_t)}{c_1}$$

$$= \exp\left\{-\frac{1}{2}\int_0^t (y_\tau - h(x_\tau, \tau))^{\mathrm{T}} R^{-1}(\tau)(y_\tau - h(x_\tau, \tau))\, d\tau\right\}. \quad (5.39)$$

In a completely analogous way, we get

$$p[x_\tau, \tau \in [0, t]] = p(x_0) \exp\left\{-\frac{1}{2}\int_0^t (dx_\tau/d\tau - f(x_\tau, \tau))^{\mathrm{T}} (GQG^{\mathrm{T}})^{-1} \times (dx_\tau/d\tau - f(x_\tau, \tau))\, d\tau\right\}. \quad (5.40)$$

It is easy to see that maximization of the product

$$p[y_\tau, \tau \in [0, t] | x_\tau, \tau \in [0, t]]\, p[x_\tau, \tau \in [0, t]]$$

with respect to x_τ, $0 \leqslant \tau \leqslant t$ is equivalent to the continuous least squares problem (5.24), (5.25).

Estimation based on the maximization of the conditional probability density function (5.28) may be called *joint maximum likelihood (Bayesian) estimation.* The estimate is the peak or *mode* of the joint conditional density (5.28), and as such is called the maximum likelihood or most probable estimate. It is obtained by maximizing the *joint* density function (5.28) rather than the marginal density

$$p(x_N \mid y_1, \ldots, y_N). \quad (5.41)$$

The *maximum likelihood (Bayesian) estimate* $x_N{}^*$ obtained by maximizing the marginal density (5.41) is, in general, not the same as the *joint* maximum likelihood (Bayesian) estimate $\bar{x}_N$ obtained by maximizing (5.28). The optimizing sequence $\{\bar{x}_0, \ldots, \bar{x}_N\}$ is sometimes called the *modal trajectory.* $x_N{}^*$ does not, in general, lie on the modal trajectory. (It does in linear problems.) The estimate is *Bayesian*, since it is obtained by maximizing an *a posteriori* density that includes the prior data $p(x_0)$ via Bayes' rule. A non-Bayesian joint maximum likelihood estimate may be obtained by maximizing

$$p(y_1, \ldots, y_N \mid x_0, \ldots, x_N). \quad (5.42)$$

Compare this with (5.29).

Consider the maximum likelihood (Bayesian) estimate obtained by maximizing (5.41) or, in the continuous case, by maximizing

$$p(x, t \mid Y_t). \tag{5.43}$$

If these densities are Gaussian (linear problems), or just symmetric and unimodal, then their (respective) modes coincide with their (respective) means. In that case, the *maximum likelihood (Bayesian) estimate is the same as the minimum variance estimate* defined in Section 2. We note that maximum likelihood estimation is of questionable value unless the density function is unimodal and concentrated near the mode.

Many investigators have studied the least squares estimation problem in the form in which we define it, and in other related and equivalent forms. The objective of most of these studies is the development of recursive least squares filters and predictors. Some of these authors give the probabilistic interpretation of least squares which we have outlined. It is impossible to develop all these works here. Furthermore, they are not central to the theme of this book, which is the probabilistic approach to filtering and prediction. As was already noted, much of this work simply rederives probabilistic results. In the next paragraphs, we cite some of the recent least squares work. The interested reader may go to the references.

Perhaps the earliest work in discrete linear recursive least squares is that of Swerling [39]. His work predates Kalman's [19] probabilistic results, but considers only the case in which the dynamical system is noise free ($w_k \equiv 0$). Following the publication of Kalman's work, a number of authors rederive his (linear) results. These include Ho [18], Battin [2], Schmidt [33], and Mowery [28]. Swerling [40, 41] gives a modern treatment of least squares estimation. A theoretical treatment of recursive least squares is available in Albert and Sittler [1]. Cox [11, 12] studies the nonlinear discrete problem via dynamic programming and variational techniques, rederives the linear results, and gives an approximate solution for the nonlinear problem. He also gives the linear smoothing solution. Rauch *et al.* [32] rederive Kalman's result by maximizing (5.41). They also give the linear smoothing solution. Cox [14] proposes a dynamic programming formulation for a minimax problem (minimizing maximum errors, rather than their squares). Larson and Peschon [25] formulate a general nonlinear problem with non-Gaussian errors via dynamic programming. The last problem is also studied by Friedland and Bernstein [17] via variational techniques.

Kalman and Bucy's [20] continuous (linear) filtering and prediction results are rederived by Bryson and Frazier [7] using variational

techniques. They also give the linear smoothing solution. Cox [13] studies the nonlinear continuous problem via dynamic programming, rederives the linear results, and gives an approximate solution for the nonlinear problem. He also gives the linear smoothing solution, as do Rauch *et al.* [32]. The continuous nonlinear least squares estimation problem is also studied by Bellman *et al.* [3, 4] using the quasi-linearization technique. That same problem is also studied by Bellman *et al.* [5], and by Detchmendy and Sridhar [15], who, using the technique of invariant imbedding, give an approximate solution for the filtering problem. Such an approximate solution is also obtained by Friedland and Bernstein [17] using variational techniques.

The statistical methods, without a probabilistic interpretation, do not involve stochastic differential equations, and as a result are conceptually and theoretically simpler than the probabilistic approach. There is no need to make any assumptions about the noise; the problem is one of minimizing deterministic errors. On the other hand, results obtained via statistical methods are more difficult to interpret. Weighting matrices (in the least squares problems) are arbitrary, and the meaning and significance of the results are not clear. Once a probabilistic interpretation of the statistical methods is made, these methods are formal approaches to the probabilistic problem, without a rigorous mathematical basis.

4. FOREWORD AND DISCUSSION

This book develops the probabilistic approach to filtering and prediction problems. Some of the statistical methods will be pursued in examples, but are of secondary interest here. The smoothing problem is not treated in detail. Only the linear smoothers are derived in Examples 7.8 and 7.16, and the interested reader can consult the references cited in Section 1 for additional material on smoothing.

There are several ways in which our definitions of the estimation problems can be generalized. We have assumed that the system noise (w) and measurement noise (v) are independent. More generally, these could be assumed correlated. This case is treated by Kalman [21], Cox [13], Friedland and Bernstein [17], and Kushner [23].[8] We have also assumed that all the noise inputs are white (uncorrelated in time). The case of colored (time-correlated) system noise can often be reduced to the white noise case by state-augmentation (Section 4.8), and therefore our white noise assumption represents little loss of generality. See

[8] Kushner's paper contains a small algebraic error in terms involving this correlation.

Section 4.8 on modeling, and also Section 4.10. A similar procedure in case of colored measurement noise leads to a singular covariance matrix (R), which is also the case when some of the observations are noise-free. This presents no particular theoretical difficulties in discrete problems (see Kalman [21]), but is essential in the continuous case. In this connection, additional results for discrete problems are contained in Cox [12] and Bryson and Henrikson [9]. For the continuous case, see Bryson and Johansen [8], Cox [13], Mehra and Bryson [27], and Bucy [10]. Some of the generalizations we have noted, although they do not appear in the main body of the book, will be treated in examples.

Finally, we note that our work is restricted to dynamical systems in which the state is a continuous or discrete parameter, continuous state space (Markov) process. Nonlinear filtering results for continuous parameter Markov chains are available in Wonham [44] and Shiryaev [36].

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